
DR BEN MASLEN

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PROFESSIONAL SUMMARY

Interdisciplinary statistician and data scientist with a PhD in Statistics from the University of New South Wales and CSIRO Data61. I develop statistical and artificial intelligence methods that help ecologists monitor and protect the natural environment, with expertise in ecological statistics, environmental sensing, computer vision, and applied data science. I have a passion for teaching and communicating statistics to diverse audiences.

EXPERIENCE

LECTURER, 2026, JUL - CURRENT
SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW SOUTH WALES

Recently lectured and co-convened the statistics component of Civil and Environmental Engineering Computations (CVEN2002), responsible for delivering and coordinating statistics teaching for over 450 second-year engineering students, and I will be lecturing Statistics for Life and Social Sciences (MATH1041) in T3 of 2026. I am also conducting my own research at the intersection of ecology, statistics, and data science, as well as interdisciplinary applied research through ongoing collaborations with other researchers at UNSW.

STATISTICAL CONSULTANT, 2021, JUL - 2026, JUN &
MARK WAINWRIGHT ANALYTICAL CENTRE, UNIVERSITY OF NEW SOUTH WALES 2018, SEP - 2019, DEC

Collaborated with over 220 academics, researchers, and higher degree research students at the University of New South Wales and partner institutions from a variety of disciplines to provide appropriate design and analysis advice relative to their research questions and data properties. I also regularly prepared content and facilitated seminars and courses on coding and statistical analysis methods for researchers and higher degree research students. Through this role, I was contracted by the School of Mathematics and Statistics to supervise multiple master's student groups taking the Data Science and Decisions Project course (DATA5925 & MATH5925), guiding students through an independent research project tackling real world data science problems (2026).

DATA SCIENTIST, 2026, JAN - 2026, FEB,
WENTWORTH GROUP OF CONCERNED SCIENTISTS 2020, JAN - 2020, JUN

Applied statistical analysis to environmental policy questions for the Wentworth Group of Concerned Scientists, investigating changes in the outward flow and reliability of water shares under a drier climate in the Murray-Darling Basin and land clearing in NSW following 2016 environmental reforms.

PHD IN AI AND ECOLOGICAL STATISTICS, 2021, JUNE - 2025, AUG
UNIVERSITY OF NEW SOUTH WALES & CSIRO DATA6

During my PhD, I developed and applied advanced AI techniques and statistical modelling to improve the analysis of ecological video data. I enhanced AI object detection by incorporating animal movement patterns, evaluated the reliability of commonly used ecological metrics, and implemented simulation-based approaches to assess bias and statistical power. I also adapted distance sampling methods to estimate fish density, enabling a novel approach to gain accurate ecological insights from imperfect observational data.

STATISTICAL INSIGHTS ANALYST, 2020, SEP - 2021, DEC
HEALTHDIRECT

Developed reporting and advanced modelling insights to enhance the management of health and user experience data across a range of Healthdirect services.

DATA ANALYST, 2018, MAY - 2019, MAY
NATURAL RESOURCE ACCESS REGULATOR, NSW GOVERNMENT

Developed and implemented a reporting and data management system for the NSW Natural Resource Access Regulator. This enabled the tracking of workflow for the strategic management and triaging of tasks.

MATHEMATICS AND STATISTICS DEMONSTRATOR, 2018, FEB - 2018, MAY
SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW SOUTH WALES

Academic demonstrator for various mathematics and statistics courses including; Higher Probability and Stochastic Processes, Numerical Methods and Statistics, and Maths for Life and Social Sciences.

<https://BenMaslen.github.io>

EDUCATION

PHD IN STATISTICS, UNIVERSITY OF NEW SOUTH WALES & CSIRO DATA61	2021-2025
HONOURS OF STATISTICS (1 ST CLASS), UNIVERSITY OF NEW SOUTH WALES	2017
BACHELOR OF ADVANCED SCIENCE (HIGH DISTINCTION), UNIVERSITY OF NEW SOUTH WALES	2014-2016

TEACHING EXPERTISE

LECTURING:

• Civil and Environmental Engineering Computations, (School of Mathematics and Statistics, UNSW)	2026
• Introduction to Mixed Models with R (Stats Central, UNSW)	2026
• Statistics for Medical Science (Stats Central, UNSW)	2025-2026
• Introduction to Regression with R (Stats Central, UNSW)	2019, 2021-2025
• Introduction to R (Stats Central, UNSW)	2019, 2022
• Introduction to Statistics with R (Stats Central, UNSW)	2022
• Ecological statistics (Stats Central, UNSW)	2018-2019, 2021-2022

ACADEMIC DEMONSTRATOR:

• Higher Probability and Stochastic Processes (School of Mathematics and Statistics, UNSW)	2018
• Numerical Methods and Statistics (School of Mathematics and Statistics, UNSW)	2018
• Mathematics for Life and Social Sciences (School of Mathematics and Statistics, UNSW)	2018

CASUAL MARKER:

• Statistical Inference (School of Mathematics and Statistics, UNSW)	2023-2025
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SUPERVISION:

School of Mathematics and Statistics, UNSW (2026) – Supervisor for two groups of Masters students taking the Data Science and Decisions Project course (DATA5925 & MATH5925), guiding students to independently research and analyse a VISA tribunal outcomes dataset, tackling real world data science problems.

Stats Central, UNSW (2018-2019, 2021-2026) – I have met and helped with the supervision of over 180 honours and postgraduate research students, guiding them through the experimental design and analysis phase of their research projects in collaboration with their academic supervisors.

TECHNICAL AND CODING SKILLS

• R (Statistical Computing Language)	• SPSS (Statistical Computing Software)
• Python (General Purpose Programming Language)	• Matlab (Technical Computing Language)
• SQL (Data Management Language)	• Bash/Linux (High Performance Computing)

PROFESSIONAL DEVELOPMENT CERTIFICATIONS

• FOUNDATIONS OF DATA SCIENCE: COMPUTATIONAL THINKING WITH PYTHON	2020
• University of California, Berkeley	

PROFESSIONAL MEMBERSHIPS

- STATISTICAL SOCIETY OF AUSTRALIA (**SSA**)
- AUSTRALIAN SOCIETY FOR FISH BIOLOGY (**ASFB**)

PUBLICATION RECORD

Peer reviewed research articles published in Q1 journals: 12

Peer reviewed research articles published in Q2 journals: 4

Total citations: 320

H index: 8

i10 index: 7

Publications of interest:

The Motion Picture: Leveraging Movement to Enhance AI Object Detection in Ecology 2025

This study evaluated frame differencing, background subtraction, optical flow and tracking methods integrated into a YOLOv8 AI object-detection pipeline in Python. Across more than 35,000 annotated images from terrestrial, marine and freshwater habitats, we found movement information was especially valuable for smaller studies and rarer species. The implementation is available through a public [GitHub](#) repository, with a public benchmark dataset released through [Dryad](#).

How many sites? Methods to assist design decisions when collecting multivariate data in ecology 2023

This paper developed a power-analysis framework for multivariate ecological data combining Gaussian copula and factor-analytic models, expert-informed effect sizes and an efficient critical-value approach to estimate power. The accompanying open-source [ecopower](#) R package makes the method accessible to ecologists and has been downloaded more than 14,000 times from CRAN.

ACHIEVEMENTS AND PRESENTATIONS

Invited Speaker:

- Australian Statistical Conference, Perth, Australia 2025
 - Mentor for statistical consulting session

Contributed Speaker:

- Australian Fish Biology Conference, Darwin, Australia 2025
- International Statistical Ecology Conference, Swansea, Wales 2024
- UNSW Mathematics and Statistics Postgraduate Conference, Sydney, Australia 2023, 2024
 - Peoples choice award for best presentation (2024)
- UNSW 1 minute thesis competition 2022
 - 1st place in the School of Mathematics and Statistics
- Statistical Ecology and Environmental Monitoring Conference, Queenstown, New Zealand 2017

Scholarships:

- Australian Government Research Training program scholarship 2021-2024
- CSIRO, Data61 PHD top up scholarship 2021-2024
- UNSW School of Mathematics and Statistics PHD top up scholarship 2021-2024

Awards:

- Tjanpi Award for best Student paper in Environmental Statistics (runner up) 2023

ACADEMIC REFEREES

- Professor David Warton (PhD supervisor & colleague at the School of Mathematics and Statistics) - david.warton@unsw.edu.au, +61 2 9385 7031
- Professor Jake Olivier (manager at the School of Mathematics and Statistics) - j.olivier@unsw.edu.au, 02 9385 6656
- Dr Nancy Briggs (manager at Stats Central) - nancy.briggs@unsw.edu.au, +61 2 9385 8220
- Dr Peter Humburg (colleague at Stats Central) - peter.humburg@unsw.edu.au, +61 2 9385 8220